Constant Temperature Humidity Test Chamber



Case type



Narrow type

(picture for reference, different structure is optional, with same performance)

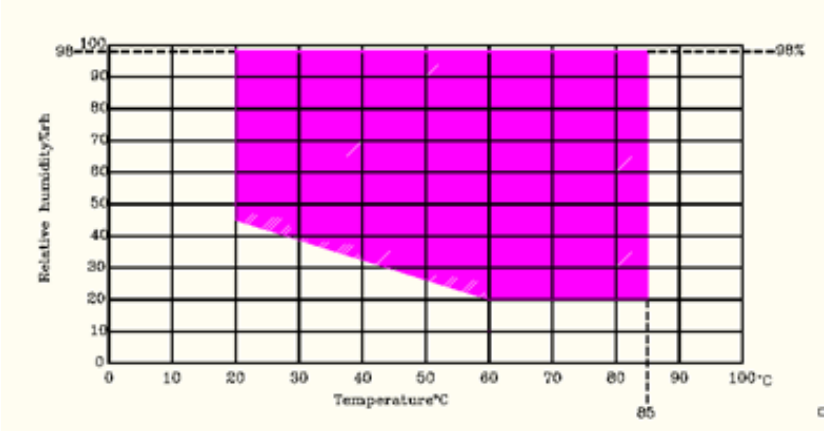
Models optional:

Model	Inner size (unit:cm) volume	Temp.	External size (unit:cm) (narrow/case) appr.	Power supply	Installation power
ZH-HS-80B	W40*D40*H50 (80L)	0~150℃	W69*D142*H157 W94*D90*H158	AC220V±10%, 50Hz, 1ph	5.0KW,23A
ZH-HS-80C		-20~150℃			
ZH-HS-80D		-40~150℃			5.5KW,25A
ZH-HS-80E		-60~150℃			
ZH-HS-80F		-70~150℃			
ZH-HS-150B	W50*D50*H60 (150L)	0~150℃	W79*D152*H167 W104*D100*H163	AC220V±10%, 50Hz, 1ph	5.0KW,23A
ZH-HS-150C		-20~150℃			
ZH-HS-150D		-40~150℃			5.5KW,25A
ZH-HS-150E		-60~150℃			
ZH-HS-150F		-70~150℃			

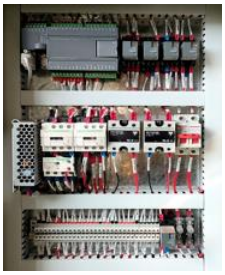
ZH-HS-225B	W50*D60*H75 (225L)	0~150℃	W79*D162*H184 W104*D111*H182	AC220V±10%, 50Hz, 1ph	9.0KW，40A
ZH-HS-225C		-20~150℃			
ZH-HS-225D		-40~150℃		AC380V±10%, 50Hz, 3ph	10.0KW, 25A
ZH-HS-225E		-60~150℃			12.5KW, 28A
ZH-HS-225F		-70~150℃			
ZH-HS-408B	W60*D80*H85 (408L)	0~150℃	W89*D183*H194 W113*D133*H194	AC220V±10%, 50Hz, 1ph	9.0KW，40A
ZH-HS-408C		-20~150℃		AC380V±10%, 50Hz, 3ph	10.0KW, 25A
ZH-HS-408D		-40~150℃			10.5KW, 26A
ZH-HS-408E		-60~150℃			13.0KW, 28A
ZH-HS-408F		-70~150℃			
ZH-HS-800B	W80*D100*H100 (800L)	0~150℃	W128*D184*H208 W153*D133*H210	AC380V±10%, 50Hz, 3ph	10.5KW, 26A
ZH-HS-800C		-20~150℃			11.0KW, 26A
ZH-HS-800D		-40~150℃			13.0KW, 28A
ZH-HS-800E		-60~150℃			14.5KW, 32A
ZH-HS-800F		-70~150℃			
ZH-HS-1000B	W100*D100*H100 (1000L)	0~150℃	W128*D204*H208 W153*D153*H210	AC380V±10%, 50Hz, 3ph	10.5KW, 26A
ZH-HS-1000C		-20~150℃			11.0KW, 26A
ZH-HS-1000D		-40~150℃			13.0KW, 28A
ZH-HS-1000E		-60~150℃			14.5KW, 32A
ZH-HS-1000F		-70~150℃			
Remark: other voltage, frequency 60Hz can be customized					

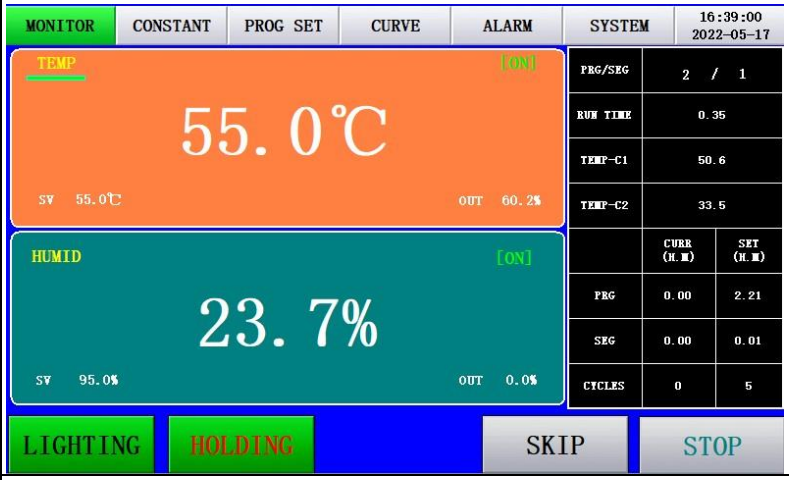
Technical specification:

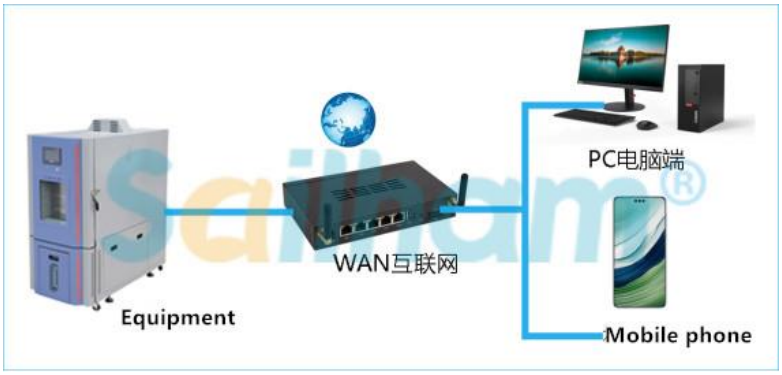
1. Item	**Constant Temperature humidity test chamber**
2. Patent	ZL201620138459.2 A constant temperature humidity test chamber
3. Function	It simulates the products in the chamber where the climate condition of different temperature and humidity, such as high-low temperature operation and storage, temperature cycling, high-low temperature and high-low humidity, moisture condensation etc. to check its adaptability and changing performance. It has different name like “climate test chamber”, “environmental test chamber”, “Damp heat test chamber” etc.. It’s required to conforms to IEC, JIS, GB, MIL to reach the consistency (including test schedule, condition, method) under the international standard.

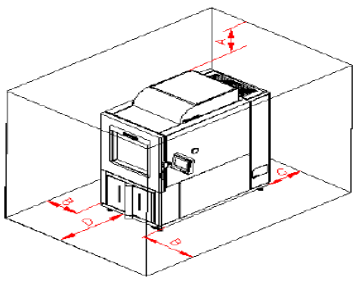
4. Sample application & prohibited	The chamber applies to: compatibility test for store, transportation and application of electric, electronic products, unit components, parts and other materials under high or low temperature / humidity; The following samples are prohibited to be tested or stored: - Inflammable , explosive and other volatile substances; - Corrosive substance; - Biont; - Strong electromagnetic emission source.
5. Capacity & Performance	
5.1. Test space	The test space can not be more than three-quarters of the total space
5.2. Temperature changing rate	Heating rate: 2.0℃-4.0℃/min; Cooling rate: 0.7℃-1.0℃/min (average non-linear, no load)
5.3. Temperature fluctuation	±0.5℃
5.4. Temperature uniformity	±2.0℃
5.5. Temperature deviation	±2.0℃
5.6. Humidity & temperature control range (refer to fig.)	**20% ~ 98%R.H (purple part is controllable)** 
5.7. Humidity fluctuation	±2.5%R.H.
5.8. Humidity deviation	±3.0%
5.9. Humidity uniformity	±3.0% (≥75%); ±5.0% (< 75%)

※ The above temperature and humidity performance is measured under the condition of ambient temperature 25℃±3℃, relative humidity ≤85%, ,no-load and no sample inside chamber.	
6. Structural features	
6.1. Chamber material	In-wall material: SUS304 External material: **painted shell or SUS304 shell optional (painted shell by default)** Insulation material: PU polyurethane foam, thickness 100mm
6.2. Duct circulation	 Centrifugal circulation motor, upper air outlet and lower return air
6.3. Test chamber	Four layers of vacuum anti-condensation glass, the viewing window size: 250*300 (mm) appr. Lighting tube inside chamber Access port / test hole: one test hole of φ50mm or φ100mm by default on the left side  
6.4. Sample holder	304#SUS Stainless steel punching stencil; Qty: 2pcs; height of shelves is adjustable 
6.5. Power line & drain	There is a condensate drain on the back of the device and a national standard 2.5m long cable.

6.6. Door	Open the single door from left to right handle; with viewing window, window lamp with condensation prevention thermal device
6.7. Control panel	7-inch touch screen controller, power switch, running indicator, USB data download interface All the history curve and history data can be downloaded from controller as Excel or JPG of curve according to date.
6.8. Electronic control system	 picture for reference. ●Strictly implement relevant standard and normative operations; ●All lines are clearly marked and easy to maintain; ●All lines are equipped with wire terminals for point-to-point connection, with high reliability and long service life.
7. Meet the technical standards	GB/T 2423.1-2001 Test A: Low temperature test method GB/T 2423.2-2001 Test B: High temperature test method GJB 150.3-1986 High temperature test GJB 150.4-1986 Low temperature test IEC68-2-1 Test A: Cold IEC68-2-2 Test B: Dry heat GB11158 “Technical Conditions for High Temperature Test Chamber ” GB10586-1 “Technical conditions for Damp heat test chamber ” GB/T 2423.2 “Basic environmental testing procedures for electric and electronic products B: High temperature test method”. GB/T 2423.3 “Basic environmental testing procedures for electric and electronic products Ca: Constant damp heat testing method” etc.
8. Refrigeration system	

8.1. Refrigeration compressor	  Sanyo (Japan) Hitachi (Japan)
8.2. Evaporator	Finned tube evaporator 
8.3. Condenser	Air-cooled condenser (Control the ambient temperature at +5～25℃± 3℃) 
8.4. Expansion system	Refrigeration system of volume control (capillary)
8.5. Refrigerant	R404a (Eco-friendly, Ozone loss rate at 0) 
9. Controller	
9.1. Control panel	7 inch touch screen controller, power switch, running indicator, USB data download interface 
9.2. Control mode	● Program mode: 50 groups of programs, each group of programs with 50 segments, can set the whole cycle of the program, part of the cycle, program connection, set the program end action and standby action. ● FIX mode: The running time and running conditions can be set.

9.3. Features	● The hardware architecture adopts: PLC listed company Xinjie + MCGS 7-inch human-machine interface touch screen ● The software is independently developed and has a software copyright registration certificate. (with Strong scalability, non-standard custom software can be modified as required) ● Flat design, convenient operation and large storage space
9.4. Sensor	PT100 three-wire high-precision sensor
9.5. Analog output	Voltage pulse (SSR) 2 points, DC 24V, minimum pulse width 5ms.
9.6. Communication port	●USB: historical data can be downloaded and saved in Excel format by date. The curve data can be saved and downloaded in the form of BMP images by taking screenshots, and can be zoomed in or out as needed before downloading screenshots.  ●RS485 communication interface (providing communication protocol, excluding software development)  ●RJ45 network port (provides communication protocol, does not include software development, supports LAN local area network connection, optional purchase for WAN external network remote connection)

10. Safety protection device	
10.1. Refrigeration system	● Compressor over-current protection ● Compressor over-pressure protection ● Compressor over-temperature protection
10.2. Electronic control system	● Heater & humidifier over temperature protection ● Overload short circuit leakage protection ● Motor overheating protection ● Water shortage protection
11. Accessories & parts	The chamber is to be transported as a unitary; Accessories as follows: ❖ Instruction Manual & warranty card; ❖ 1 set of φ50 mm or φ100 mm test hole (installed); ❖ 1 bag (10 pieces) of wet bulb gauze; ❖ 2 shelves ; ❖ castor x 4 & Foot Cup x 4 (installed).
12. Usage requirements	Assured by users as below requirements
12.1. Installation position	- With leveling ground and good ventilation; - No strong shock near the chamber; - No strong electromagnetic field near the chamber - No inflammable , explosive , corrosive material and dust - Surrounding size for chamber operation and maintenance: A: not less than 10 cm B: not less than 60 cm C: not less than 60 cm D: 225 type: not less than 80 cm 408 type: not less than 90 cm 1000 type: not less than 120 cm 
12.2. Condition	Temperature: 5℃ ~ 30℃ Humidity: ≤85% Air pressure: 86 kPa ~ 106 kPa

12.3. Water supply	Pure water or distilled water is added into the water tank manually. (Customization for water auto-supply device is available)
12.4. Power supply condition	❖ Frequency: 50 Hz (customization for 60 Hz is available) ❖ Resistance of protection grounding wire less than 4Ω. ❖ It's required to equipped with an air switch of corresponding capacities, and this switch must be used independently for the equipment (can not to be connected to a socket).

15. Core parts of the chamber

	Parts	Brand	Brand place	No.	Parts	Brand	Original
	Temperature control software	Sailham	China	14	PLC controller	XINJE	China
	Compressor	Sanyo or Hitachi	Japan	15	Touch screen	MCGS	China
	Refrigerant	DuPont	USA	16	Solid-state relay	Carlo Gavazzi	Switzerland
	Drying agent	Emerson	USA	17	AC contactor	Schneider	France
	Pressure senor	Sagomiya	Japan	18	Intermediate relay	Schneider	France
	Electronic expansion valve	Sagomiya	Japan	19	Overload protector	Schneider	France
	Gas-liquid separator	FASIKE	China	20	Switch power supply	Mean well	Taiwan/CN
	Condenser fan	MaEr	China	21	Breaker	DELIXI	China
	Refrigerant oil	Emkarate	USA	22	Main switch	DELIXI	China

	Over.temp protector	Rainbow	Korea	23	Heating tube	FeiYang	China
	Sample over.temp protector	Sailham	China	24	Humidifier tube	FeiYang	China
	Evaporator	Jiaheng	China	25	Circulating motor	Yizheng	China
	Condenser	Jiaheng	China	26	Temp/humid. sensor	Minyang	Taiwan/CN
	Water pump	HG	China		Voltage transformer	EEIO	China

Remark: Some accessories, if out of stock due to material shortage, may be replaced by brands of the same level to avoid affecting delivery time. Please be informed without further notice!